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(54) NON ORIENTED ELECTRICAL STEEL SHEET, IRON CORE, MANUFACTURING METHOD OF IRON CORE, MOTOR, AND MANUFACTURING METHOD OF MOTOR

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(57) **ABSTRACT**

A non oriented electrical steel sheet includes, as a chemical composition, by mass %, 1.0% or more and 5.0% or less of Si, wherein a sheet thickness is 0.10 mm or more and 0.35 mm or less, an average grain size is 30 μm or more and 200 μm or less, an X value defined by $X=(2\times B_{50L}+B_{50C})/(3\times I_S)$ is 0.800 or more, and an iron loss $W_{10/1k}$ is 80 W/kg or less.

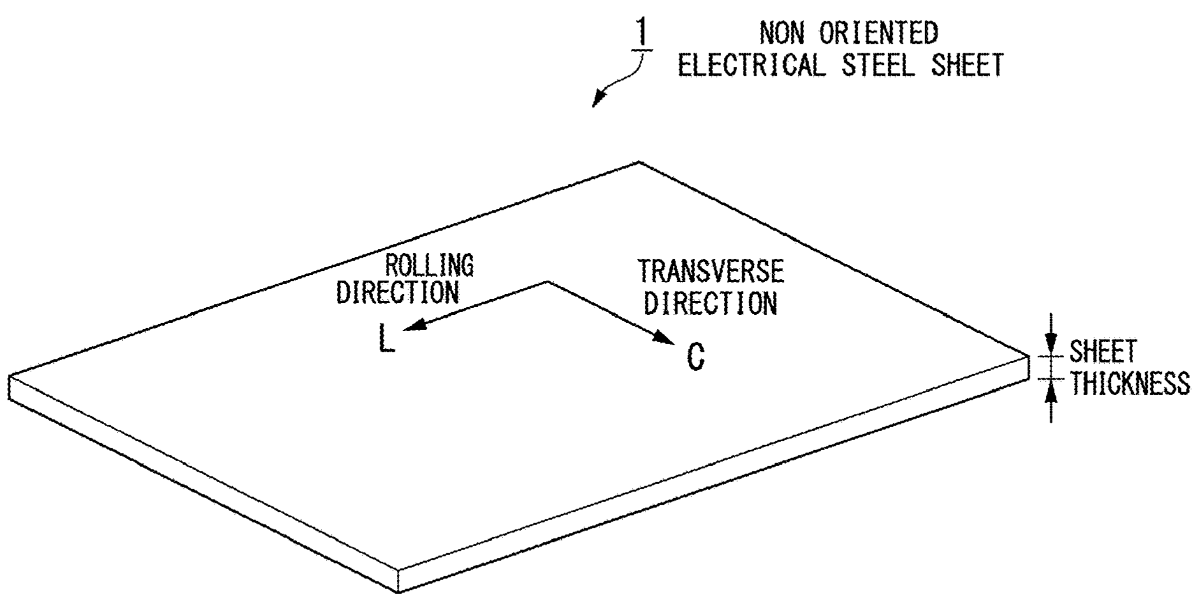
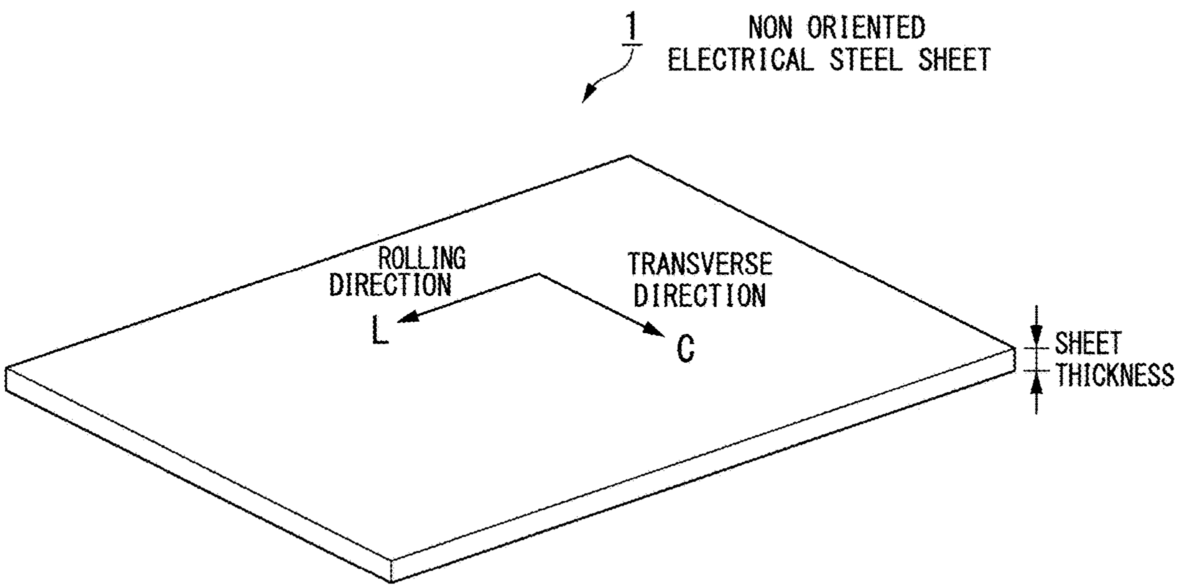


FIG. 1


1
NON ORIENTED
ELECTRICAL STEEL SHEET
ROLLING
DIRECTION
L
TRANSVERSE
DIRECTION
C
SHEET
THICKNESS

NON ORIENTED ELECTRICAL STEEL SHEET, IRON CORE, MANUFACTURING METHOD OF IRON CORE, MOTOR, AND MANUFACTURING METHOD OF MOTOR

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a Continuation of copending application Ser. No. 18/464,879, filed on Sep. 11, 2023, which is Continuation of application Ser. No. 18/077,789, filed on Dec. 8, 2022 (now U.S. Pat. No. 11,814,710, issued on Nov. 14, 2023), which is a Continuation of PCT International Application No. PCT/JP2022/029067, filed on Jul. 28, 2022, which claims priority under 35 U.S.C. 119 (a) to Patent Application No. 2021-126289, filed in Japan on Jul. 30, 2021, all of which are hereby expressly incorporated by reference into the present application.

TECHNICAL FIELD

[0002] The present invention relates to a non oriented electrical steel sheet. More specifically, the present invention relates to a non oriented electrical steel sheet, which is suitable for integrally punched iron cores of motors for electric vehicles, hybrid vehicles, or the like, an iron core, a manufacturing method of the iron core, a motor, and a manufacturing method of the motor.

[0003] Priority is claimed on Japanese Patent Application No. 2021-126289 filed on Jul. 30, 2021, the content of which is incorporated herein by reference.

BACKGROUND ART

[0004] Due to a need to reduce global warming gases, products with less energy consumption have been developed in industrial fields. For instance, in a field of an automobile, there are fuel-efficient vehicles such as hybrid-driven vehicles that combine a gasoline engine and a motor, and motor-driven electric vehicles. A technology common to these fuel-efficient vehicles is a motor, and increasing an efficiency of the motor has become an important technology.

[0005] In General, a motor includes a stator and a rotor. The stator includes an iron core, and the iron core is classified as an integrally punched iron core and a segmented iron core. For the integrally punched iron core and the segmented iron core, there is a demand for a non oriented electrical steel sheet having excellent magnetic characteristics in a rolling direction (hereinafter referred to as "L direction") and in a transverse direction (hereinafter referred to as "C direction").

[0006] In addition, the motor shows excellent performance in a case where a gap between the stator and the rotor becomes smaller as an internal structure of the motor. Thus, each component of the motor is required to have a high shape accuracy. For instance, the integrally punched iron core and the segmented iron core are both formed by punching a steel sheet blank. However, in the integrally punched iron core, since the steel sheet blank is punched to be a hollow disc shape, the shape accuracy after punching may be deteriorated due to mechanical anisotropy of the steel sheet blank. Therefore, for the integrally punched iron core, it is desired for the non oriented electrical steel sheet to have small mechanical anisotropy.

[0007] For instance, Patent Document 1 discloses a technique related to a non oriented electrical steel sheet having excellent magnetic characteristics. Patent Document 2 discloses a technique related to a non oriented electrical steel sheet that can improve efficiency of a motor including a segmented iron core. Patent Document 3 discloses a technique related to a non oriented electrical steel sheet having excellent magnetic characteristics.

RELATED ART DOCUMENTS

Patent Documents

[0008] [Patent Document 1] Japanese Patent (Granted) Publication No. 5447167

[0009] [Patent Document 2] Japanese Patent (Granted) Publication No. 5716315

[0010] [Patent Document 3] PCT International Publication No. WO2013/069754

SUMMARY OF INVENTION

Technical Problem to be Solved

[0011] The present invention has been made in consideration of the above mentioned situations. An object of the invention is to provide a non oriented electrical steel sheet with excellent magnetic characteristics and small mechanical anisotropy for an integrally punched iron core, an iron core, a manufacturing method of the iron core, a motor, and a manufacturing method of the motor.

Solution to Problem

[0012] An aspect of the present invention employs the following.

[0013] (1) A non oriented electrical steel sheet according to an aspect of the present invention includes a chemical composition containing, by mass %,

[0014] 0.005% or less of C,
[0015] 1.0% or more and 5.0% or less of Si,
[0016] less than 2.5% of sol. Al,
[0017] 3.0% or less of Mn,
[0018] 0.3% or less of P,
[0019] 0.01% or less of S,
[0020] 0.01% or less of N,
[0021] 0.10% or less of B,
[0022] 0.10% or less of O,
[0023] 0.10% or less of Mg,
[0024] 0.01% or less of Ca,
[0025] 0.10% or less of Ti,
[0026] 0.10% or less of V,
[0027] 5.0% or less of Cr,
[0028] 5.0% or less of Ni,
[0029] 5.0% or less of Cu,
[0030] 0.10% or less of Zr,
[0031] 0.10% or less of Sn,
[0032] 0.10% or less of Sb,
[0033] 0.10% or less of Ce,
[0034] 0.10% or less of Nd,
[0035] 0.10% or less of Bi,
[0036] 0.10% or less of W,
[0037] 0.10% or less of Mo,
[0038] 0.10% or less of Nb,
[0039] 0.10% or less of Y, and
[0040] a balance consisting of Fe and impurities, wherein

[0041] a sheet thickness is 0.10 mm or more and 0.35 mm or less,

[0042] an average grain size is 30 μm or more and 200 μm or less,

[0043] an X value defined by the following expression 1 is 0.800 or more, and

[0044] an iron loss $W_{10/1k}$ when excited so as to be a magnetic flux density of 1.0 T at a frequency of 1 kHz is 80 W/kg or less,

[0045] where the expression 1 is $X=(2\times B_{50L}+B_{50C})/(3\times I_S)$ and

[0046] where B_{50L} denotes a magnetic flux density in a rolling direction when magnetized with a magnetizing force of 5000 A/m, B_{50C} denotes a magnetic flux density in a transverse direction when magnetized with a magnetizing force of 5000 A/m, and I_S denotes a spontaneous magnetization at room temperature.

[0047] (2) In the non oriented electrical steel according to (1), the chemical composition may include, by mass %, more than 3.25% and 5.0% or less of Si.

[0048] (3) In the non oriented electrical steel according to (1) or (2), the chemical composition may include, by mass %, at least one of

[0049] 0.0010% or more and 0.005% or less of C,

[0050] 0.10% or more and less than 2.5% of sol. Al,

[0051] 0.0010% or more and 3.0% or less of Mn,

[0052] 0.0010% or more and 0.3% or less of P,

[0053] 0.0001% or more and 0.01% or less of S,

[0054] 0.0015% or more and 0.01% or less of N,

[0055] 0.0001% or more and 0.10% or less of B,

[0056] 0.0001% or more and 0.10% or less of O,

[0057] 0.0001% or more and 0.10% or less of Mg,

[0058] 0.0003% or more and 0.01% or less of Ca,

[0059] 0.0001% or more and 0.10% or less of Ti,

[0060] 0.0001% or more and 0.10% or less of V,

[0061] 0.0010% or more and 5.0% or less of Cr,

[0062] 0.0010% or more and 5.0% or less of Ni,

[0063] 0.0010% or more and 5.0% or less of Cu,

[0064] 0.0002% or more and 0.10% or less of Zr,

[0065] 0.0010% or more and 0.10% or less of Sn,

[0066] 0.0010% or more and 0.10% or less of Sb,

[0067] 0.001% or more and 0.10% or less of Ce,

[0068] 0.002% or more and 0.10% or less of Nd,

[0069] 0.002% or more and 0.10% or less of Bi,

[0070] 0.002% or more and 0.10% or less of W,

[0071] 0.002% or more and 0.10% or less of Mo,

[0072] 0.0001% or more and 0.10% or less of Nb, and

[0073] 0.002% or more and 0.10% or less of Y.

[0074] (4) In the non oriented electrical steel according to any one of (1) to (3), the chemical composition may include, by mass %, more than 4.0% in total of Si and sol. Al.

[0075] (5) In the non oriented electrical steel according to any one of (1) to (4), the X value may be 0.800 or more and less than 0.845.

[0076] (6) In the non oriented electrical steel according to any one of (1) to (5), the X value may be 0.800 or more and less than 0.830.

[0077] (7) An iron core according to an aspect of the present invention may include the non oriented electrical steel sheet according to any one of (1) to (6).

[0078] (8) A manufacturing method of an iron core according to an aspect of the present invention may include a process of punching and laminating the non oriented electrical steel sheet according to any one of (1) to (6).

[0079] (9) A motor according to an aspect of the present invention may include the iron core according to (7).

[0080] (10) A manufacturing method of a motor may include

[0081] a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to any one of (1) to (6) and

[0082] a process of assembling the motor using the iron core.

Effects of Invention

[0083] According to the above aspects of the present invention, it is possible to provide the non oriented electrical steel sheet with excellent magnetic characteristics and small mechanical anisotropy for the integrally punched iron core, the iron core, the manufacturing method of the iron core, the motor, and the manufacturing method of the motor.

BRIEF DESCRIPTION OF DRAWINGS

[0084] FIG. **1** is a schematic illustration of a non oriented electrical steel sheet according to an embodiment of the present invention.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

[0085] Hereinafter, a preferable embodiment of the present invention is described in detail. However, the present invention is not limited only to the configuration which is disclosed in the embodiment, and various modifications are possible without departing from the aspect of the present invention. In addition, the limitation range as described below includes a lower limit and an upper limit thereof. However, the value expressed by "more than" or "less than" does not include in the limitation range. "%" of the amount of respective elements expresses "mass %".

[0086] FIG. **1** shows a schematic illustration of a non oriented electrical steel sheet according to the embodiment of the present invention.

(Chemical Composition)

[0087] Limitation reasons of the chemical composition of the non oriented electrical steel sheet according to the present embodiment are explained.

[0088] As a chemical composition, the non oriented electrical steel sheet according to the present embodiment contains Si, optional elements as necessary, and a balance consisting of Fe and impurities. Hereinafter, each element is explained.

[0089] C: 0% or more and 0.005% or less

[0090] C (carbon) is an element contained as an impurity and deteriorates the magnetic characteristics. Thus, the C content is to be 0.005% or less. Preferably, the C content is 0.003% or less. Since it is preferable that the C content is low, a lower limit does not need to be limited, and the lower limit may be 0%. However, it is not easy to industrially control the content to be 0%, and thus, the lower limit may be more than 0% or 0.0010%.

[0091] Si: 1.0% or more and 5.0% or less

[0092] Si (silicon) is an element that is effective in increasing electrical resistivity of the steel sheet and reducing iron loss. Thus, the Si content is to be 1.0% or more. Moreover, Si is an effective element for the non oriented electrical steel sheet for the integrally punched iron core to achieve both magnetic characteristics and mechanical anisotropy. In this case, the Si content is preferably more than 3.25%, more preferably 3.27% or more, further more preferably 3.30% or more, and further more preferably 3.40% or more. On the other hand, when the Si content is excessive, a magnetic flux density deteriorates significantly. Thus, the Si content is to be 5.0% or less. The Si content is preferably 4.0% or less, and more preferably 3.5% or less.

[0093] sol. Al: 0% or more and less than 2.5%

[0094] Al (aluminum) is an optional element that is effective in increasing the electrical resistivity of the steel sheet and reducing the iron loss. However, when the content is excessive, the magnetic flux density deteriorates significantly. Thus, the sol. Al content is to be less than 2.5%. A lower limit of sol. Al does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the sol. Al content is preferably 0.10% or more. Herein, the sol. Al expresses acid-soluble aluminum.

[0095] Moreover, Si and Al are elements effective in achieving both magnetic characteristics and mechanical anisotropy. Thus, the total amount of Si and sol. Al is preferably more than 4.0%, more preferably more than 4.10%, and further more preferably more than 4.15%. On the other hand, Si and Al have a strong effect on solid solution strengthening. When the content is excessive, cold rolling becomes difficult to be performed. Thus, the total amount of Si and sol. Al is preferably less than 5.5%.

[0096] Mn: 0% or more and 3.0% or less

[0097] Mn (manganese) is an optional element that is effective in increasing the electrical resistivity of the steel sheet and reducing the iron loss. However, since an alloying cost of is higher than Si or Al, an increase in the Mn content is economically disadvantageous. Thus, the Mn content is to be 3.0% or less. Preferably, the Mn content is 2.5% or less. A lower limit of Mn does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Mn content is preferably 0.0010% or more, and more preferably 0.010% or more.

[0098] P: 0% or more and 0.3% or less

[0099] P (phosphorus) is an element generally contained as an impurity. However, P has an effect in improving texture of the non oriented electrical steel sheet and thereby improving the magnetic characteristics. Thus, P may be included as necessary. However, P is a solid solution strengthening element. When the P content is excessive, the steel sheet is hardened and thereby the cold rolling becomes difficult to be performed. Thus, the P content is to be 0.3% or less. The P content is preferably 0.2% or less. A lower limit of P does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the P content is preferably 0.0010% or more, and more preferably 0.015% or more.

[0100] S: 0% or more and 0.01% or less

[0101] S (sulfur) is contained as an impurity and forms fine MnS by bonding to Mn in steel. As a result, S suppresses grain growth during annealing and deteriorates the magnetic characteristics of the non oriented electrical steel sheet. Thus, the S content is to be 0.01% or less. The S content is preferably 0.005% or less, and more preferably 0.003% or less. Since it is preferable that the S content is low, a lower limit does not need to be limited, and the lower limit may be 0%. However, it is not easy to industrially control the content to be 0%, and thus, the lower limit may be 0.0001%.

[0102] N: 0% or more and 0.01% or less

[0103] N (nitrogen) is contained as an impurity and forms fine AlN by bonding to Al in steel. As a result, N suppresses the grain growth during annealing and deteriorates the magnetic characteristics. Thus, the N content is to be 0.01% or less. The N content is preferably 0.005% or less, and more preferably 0.003% or less. Since it is preferable that the N content is low, a lower limit does not need to be limited, and the lower limit may be 0%. However, it is not easy to industrially control the content to be 0%, and thus, the lower limit may be 0.0001% or more, may be more than 0.0015%, or may be 0.0025% or more.

[0104] Sn: 0% or more and 0.10% or less

[0105] Sb: 0% or more and 0.10% or less

[0106] Sn (tin) and Sb (antimony) are optional elements having effect in improving the texture of the non oriented electrical steel sheet and thereby improving the magnetic characteristics (for instance, magnetic flux density). Thus, Sn and Sb may be included as necessary. However, when the content is excessive, the steel may become brittle and fracture may occur during cold rolling. Moreover, the magnetic characteristics are deteriorated. Thus, the Sn content and the Sb content are to be 0.10% or less, respectively. Lower limits of Sn and Sb do not need to be limited, and the lower limits may be 0%. In order to reliably obtain the above effect, the Sn content is preferably 0.0010% or more, and more preferably 0.01% or more. Moreover, the Sb content is preferably 0.0010% or more, more preferably 0.002% or more, further more preferably 0.01% or more, and further more preferably more than 0.025%.

[0107] Ca: 0% or more and 0.01% or less

[0108] Ca (calcium) is an optional element that suppresses precipitation of fine sulfides (MnS, Cu_2S, or the like) by forming coarse sulfides. When the Ca content is favorable, inclusions are controlled, the grain growth is improved during annealing, and thereby, the magnetic characteristics (for instance, iron loss) are improved. However, when the content is excessive, the effect thereof is saturated, and the cost increases. Thus, the Ca content is to be 0.01% or less. The Ca content is preferably 0.008% or less, and more preferably 0.005% or less. A lower limit of Ca does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Ca content is preferably 0.0003% or more. The Ca content is preferably 0.001% or more, and more preferably 0.003% or more.

[0109] Cr: 0% or more and 5.0% or less

[0110] Cr (chromium) is an optional element that increases the electrical resistivity and improves the magnetic characteristics (for instance, iron loss). However, when the content is excessive, a saturation magnetic flux density may decrease, the effect thereof is saturated, and the cost increases. Thus, the Cr content is to be 5.0% or less. The Cr content is preferably 0.5% or less, and more preferably 0.1% or less. A lower limit of Cr does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Cr content is preferably 0.0010% or more.

[0111] Ni: 0% or more and 5.0% or less

[0112] Ni (nickel) is an optional element that improves the magnetic characteristics (for instance, saturation magnetic flux density). However, when the content is excessive, the effect thereof is saturated, and the cost increases. Thus, the Ni content is to be 5.0% or less. The Ni content is preferably 0.5% or less, and more preferably 0.1% or less. A lower limit of Ni does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Ni content is preferably 0.0010% or more.

[0113] Cu: 0% or more and 5.0% or less

[0114] Cu (copper) is an optional element that improves strength of the steel sheet. However, when the content is excessive, the saturation magnetic flux density may decrease, the effect thereof is saturated, and the cost increases. Thus, the Cu content is to be 5.0% or less. The Cu content is preferably 0.1% or less. A lower limit of Cu does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Cu content is preferably 0.0010% or more.

[0115] Ce: 0% or more and 0.10% or less

[0116] Ce (cerium) is an optional element that suppresses the precipitation of fine sulfides (MnS, Cu_2S, or the like) by forming coarse sulfides coarse oxysulfides, or the like. As a result, the grain growth is improved, and the iron loss is improved. However, when the content is excessive, the iron loss may be deteriorated by forming oxides in addition to sulfides and oxysulfides, and the effect thereof is saturated, and the cost increases. Thus, the Ce content is to be 0.10% or less. The Ce content is preferably 0.01% or less, more preferably 0.009% or less, and further more preferably 0.008% or less. A lower limit of Ce does not need to be limited, and the lower limit may be 0%. In order to reliably obtain the above effect, the Ce content is preferably 0.001% or more. The Ce content is more preferably 0.002% or more, more preferably 0.003% or more, and further more preferably 0.005% or more.

[0117] In addition to the above elements, the non oriented electrical steel sheet according to the present embodiment may contain, as a chemical composition, the optional elements such as B, O, Mg, Ti, V, Zr, Nd, Bi, W, Mo, Nb, and Y. Amounts of these optional elements may be controlled on the basis of known knowledge. For instance, the amounts of these optional elements may be as follows.

[0118] B: 0% or more and 0.10% or less

[0119] O: 0% or more and 0.10% or less

[0120] Mg: 0% or more and 0.10% or less

[0121] Ti: 0% or more and 0.10% or less

[0122] V: 0% or more and 0.10% or less

[0123] Zr: 0% or more and 0.10% or less

[0124] Nd: 0% or more and 0.10% or less

[0125] Bi: 0% or more and 0.10% or less

[0126] W: 0% or more and 0.10% or less

[0127] Mo: 0% or more and 0.10% or less

[0128] Nb: 0% or more and 0.10% or less

[0129] Y: 0% or more and 0.10% or less

[0130] Moreover, the non oriented electrical steel sheet according to the present embodiment may contain, as a chemical composition, by mass %, at least one of

[0131] 0.0010% or more and 0.005% or less of C,

[0132] 0.10% or more and less than 2.5% of sol. Al,

[0133] 0.0010% or more and 3.0% or less of Mn,

[0134] 0.0010% or more and 0.3% or less of P,

[0135] 0.0001% or more and 0.01% or less of S,

[0136] 0.0015% or more and 0.01% or less of N,

[0137] 0.0001% or more and 0.10% or less of B,

[0138] 0.0001% or more and 0.10% or less of O,

[0139] 0.0001% or more and 0.10% or less of Mg,

[0140] 0.0003% or more and 0.01% or less of Ca,

[0141] 0.0001% or more and 0.10% or less of Ti,

[0142] 0.0001% or more and 0.10% or less of V,

[0143] 0.0010% or more and 5.0% or less of Cr,

[0144] 0.0010% or more and 5.0% or less of Ni,

[0145] 0.0010% or more and 5.0% or less of Cu,

[0146] 0.0002% or more and 0.10% or less of Zr,

[0147] 0.0010% or more and 0.10% or less of Sn,

[0148] 0.0010% or more and 0.10% or less of Sb,

[0149] 0.001% or more and 0.10% or less of Ce,

[0150] 0.002% or more and 0.10% or less of Nd,

[0151] 0.002% or more and 0.10% or less of Bi,

[0152] 0.002% or more and 0.10% or less of W,

[0153] 0.002% or more and 0.10% or less of Mo,

[0154] 0.0001% or more and 0.10% or less of Nb, and

[0155] 0.002% or more and 0.10% or less of Y.

[0156] The B content is preferably 0.01% or less, the O content is preferably 0.01% or less, the Mg content is preferably 0.005% or less, the Ti content is preferably 0.002% or less, the V content is preferably 0.002% or less, the Zr content is preferably 0.002% or less, the Nd content is preferably 0.01% or less, the Bi content is preferably 0.01% or less, the W content is preferably 0.01% or less, the Mo content is preferably 0.01% or less, the Nb content is preferably 0.002% or less, and the Y content is preferably 0.01% or less. Moreover, the Ti content is preferably 0.001% or more, the V content is preferably 0.002% or more, and the Nb content is preferably 0.002% or more.

[0157] The chemical composition as described above may be measured by typical analytical methods for the steel. For instance, the chemical composition may be measured by using ICP-AES (Inductively Coupled Plasma-Atomic Emission Spectrometer: inductively coupled plasma emission spectroscopy spectrometry). Herein, the acid soluble Al may be measured by ICP-AES using filtrate after heating and dissolving the sample in acid. In addition, C and S may be measured by the infrared absorption method after combustion, N may be measured by the thermal conductometric method after fusion in a current of inert gas, and O may be measured by, for instance, the non-dispersive infrared absorption method after fusion in a current of inert gas.

[0158] The above chemical composition is that of the non oriented electrical steel sheet without insulation coating. When the non oriented electrical steel sheet to be the measurement sample has the insulation coating and the like on the surface, the chemical composition is measured after removing the coating. For instance, the insulation coating may be removed by the following method. First, the non oriented electrical steel sheet having the insulation coating and the like is immersed in sodium hydroxide aqueous solution, sulfuric acid aqueous solution, and nitric acid aqueous solution in this order. The steel sheet after the immersion is washed. Finally, the steel sheet is dried with warm air. Thereby, it is possible to obtain the non oriented electrical steel sheet from which the insulation coating is removed. Alternatively, the insulation coating may be removed by grinding.

(Magnetic Characteristics)

[0159] As the magnetic flux density, an X value defined by the following expression 1 is to be 0.800 or more. In order to improve the magnetic characteristics, the X value is preferably 0.820 or more.

$$X = (2 \times B_{50L} + B_{50C})/(3 \times I_S) \quad \text{Expression 1}$$

[0160] Herein,

[0161] B_{50L} denotes the magnetic flux density in a rolling direction when magnetized with magnetizing force of 5000 A/m,

[0162] B_{50C} denotes the magnetic flux density in a transverse direction when magnetized with magnetizing force of 5000 A/m, and

[0163] I_S denotes spontaneous magnetization at room temperature.

[0164] I_S in the expression 1 may be obtained by the following expression 2 and expression 3. The expression 2 is for obtaining the spontaneous magnetization assuming that the spontaneous magnetization of the steel sheet is simply attenuated by elements other than Fe. Density of the steel sheet in the expression 2 may be measured on the basis of JIS Z 8807:2012. In a case where the insulation coating is applied, the density may be measured by the above described method under condition such that the insulation coating exists, and the same value of the density is also used at the time of evaluating the magnetic characteristics described later. Density of Fe in the expression 2 may be 7.873 g/cm^3.

$$I_S = 2.16 \times \{(\text{density of steel sheet})/(\text{density of Fe})\} \times [\text{Fe content (mass \%)}]/100 \quad \text{Expression 2}$$

$$\text{Fe content (mass \%)} = 100 \text{ (mass \%)} - [\text{total amount (mass \%) of C, Si, Mn, } sol. \text{ Al, P, S, N, B, O, Mg, Ca, Ti, V, Cr, Ni, Cu, Zr, Sn, Sb, Ce, Nd, Bi, W, Mo, Nb, and Y}] \quad \text{Expression 3}$$

[0165] In order to achieve both magnetic characteristics and mechanical anisotropy as the non oriented electrical steel sheet for the integrally punched iron core, the X value is preferably less than 0.845, more preferably less than 0.840, further more preferably less than 0.835, and further more preferably less than 0.830.

[0166] As the iron loss, an iron loss $W_{10/1k}$ when excited so as to have the magnetic flux density of 1.0 T at frequency of 1 kHz is to be 80 W/kg or less. The iron loss $W_{10/1k}$ is preferably 70 W/kg or less, and more preferably 49 W/kg or less. Although a lower limit of the iron loss $W_{10/1k}$ does not need to be limited, the lower limit may be 30 W/kg as necessary.

[0167] The magnetic characteristics may be measured on the basis of the single sheet tester (SST) method regulated by JIS C 2556:2015. Instead of taking a test piece with size regulated by JIS, a test piece with smaller size, for instance, a test piece of width 55 mm×length 55 mm, may be taken and measured on the basis of the single sheet tester. In a case where the test piece of width 55 mm×length 55 mm is hardly taken, the measurement based on the single sheet tester may be performed using two test pieces of width 8 mm×length 16 mm as a test piece of width 16 mm×length 16 mm. At that time, it is preferable to use an Epstein equivalent value which is converted so as to correspond to a measurement value with an Epstein tester regulated in JIS C 2550:2011.

(Average Grain Size)

[0168] When grain size is excessively coarse or fine, the iron loss under high frequency may deteriorates. Thus, the average grain size is to be 30 μm or more and 200 μm or less.

[0169] The average grain size may be measured on the basis of an intercept method regulated by JIS G 0551:2020. For instance, in a longitudinal sectional micrograph, an average value of grain sizes may be measured by the intercept method along sheet thickness direction and rolling direction. As the longitudinal sectional micrograph, an optical micrograph may be used, and for instance, a micrograph obtained at a magnification of 50-fold may be used.

(Sheet Thickness)

[0170] Sheet thickness is to be 0.35 mm or less. The sheet thickness is preferably 0.30 mm or less. On the other hand, when the sheet thickness is excessively thin, productivity of the steel sheet and motor deteriorates significantly. Thus, the sheet thickness is to be 0.10 mm or more. The sheet thickness is preferably 0.15 mm or more.

[0171] The sheet thickness may be measured by a micrometer. When the non oriented electrical steel sheet to be the measurement sample has the insulation coating and the like on the surface, the sheet thickness is measured after removing the coating. The method for removing the insulation coating is as described above.

[0172] In addition, as mechanical anisotropy, a roundness after punching to be true circle is preferably more than 0.9997 and 1.0000 or less. Specifically, when the punching is conducted using a die with a hollow disc shape of an inner diameter of 80.0 mm and an outer diameter of 100 mm and when 60 sheets of punched pieces are laminated and fastened to be formed, a value (roundness) obtained by dividing a minimum value of a diameter of an inner circumference of the formed piece by a maximum value of a diameter of the inner circumference of the formed piece is preferably more than 0.9997 and 1.0000 or less.

[0173] When the above roundness is more than 0.9997, a shape accuracy of the punched piece can be regarded as high. As a result, when it is used for the motor, an increase in cogging torque and an increase in vibration noise can be favorably suppressed. The above roundness is preferably more than 0.9998, and more preferably 0.9999 or more.

[0174] The roundness may be measured by the following method. The non oriented electrical steel sheet is punched using the die with the hollow disc shape (true circle) of the inner diameter of 80.0 mm and the outer diameter of 100 mm at a punching speed of 250 strokes/min by a 25 t continuous progressive press-working apparatus. The 60 sheets of punched pieces are laminated and fastened to form the core. The obtained ring-shaped core simulates the integrally punched iron core for the stator of the motor, and the roundness of the inner circumference can be used as an index of accuracy of an air gap with a rotor core. Diameters of the inner circumference of the obtained ring-shaped core are measured at plural positions, and a ratio of a minimum value to a maximum value of the measured diameters is regarded as the roundness. Specifically, when the punching is conducted using the die with the hollow disc shape (true circle) of the inner diameter of 80.0 mm and the outer diameter of 100 mm and when the 60 sheets of punched pieces are laminated and fastened to be formed, the value obtained by dividing the minimum value of the diameter of the inner circumference of the formed piece by the maximum value of the diameter of the inner circumference of the formed piece is regarded as the roundness.

[0175] The non oriented electrical steel sheet according to the present embodiment is excellent in both the magnetic characteristics and the roundness for the integrally punched iron core. For instance, the non oriented electrical steel sheet according to the present embodiment satisfies the X value of 0.800 or more and the iron loss $W_{10/1k}$ of 80 W/kg or less, and as a result, it is possible to obtain the effect such that the roundness is excellent. Moreover, when the chemical composition and manufacturing conditions are favorably controlled, the non oriented electrical steel sheet according to the present embodiment satisfies the X value of 0.800 or more and less than 0.845 and the iron loss $W_{10/1k}$ of 49 W/kg or less, and as a result, it is possible to obtain the effect such that the roundness is more excellent. In this case, it can be judged that the magnetic characteristics and the roundness are simultaneously and preferably achieved for the integrally punched iron core.

(Iron Core and Motor)

[0176] Since the non oriented electrical steel sheet according to the present embodiment has excellent magnetic characteristics and small mechanical anisotropy, it is suitable for the integrally punched iron core of motor for electric vehicles, hybrid vehicles, or the like. Thus, an iron core including the non oriented electrical steel sheet according to the present embodiment exhibits excellent performance. Moreover, since the non oriented electrical steel sheet according to the present embodiment is suitable for the integrally punched iron core, a motor including the iron core exhibits excellent performance.

(Manufacturing Method)

[0177] Hereinafter, an instance of a manufacturing method of the non oriented electrical steel sheet according to the present embodiment is explained below. The non oriented electrical steel sheet according to the present embodiment is not particularly limited in the manufacturing method as long as the above features are included. The following manufacturing method is an instance for manufacturing the non oriented electrical steel sheet according to the present embodiment and a favorable manufacturing method of the non oriented electrical steel sheet according to the present embodiment.

[0178] The manufacturing method of the non oriented electrical steel sheet according to the present embodiment includes the following processes (A) to (D).

[0179] (A) A first cold rolling process of subjecting a hot rolled steel sheet having the chemical composition described above to cold rolling under conditions such that a rolling reduction is 10% or larger and 75% or smaller.

[0180] (B) An intermediate annealing process of subjecting the cold rolled steel sheet obtained in the first cold rolling process to intermediate annealing under condition such that an average heating rate from 500° C. to 650° C. is 30° C./hour or faster and 1000° C./second or slower (0.0083° C./second or faster and 1000° C./second or slower), a retention temperature is 700° C. or higher and 1100° C. or lower, and a retention time is 10 seconds or longer and 40 hours or shorter (0.0028 hours or longer and 40 hours or shorter).

[0181] (C) A second cold rolling process of subjecting the intermediate-annealed steel sheet obtained in the intermediate annealing process to cold rolling under conditions such that a rolling reduction is 50% or larger and 85% or smaller to obtain a sheet thickness of 0.10 mm or more and 0.35 mm or less.

[0182] (D) A final annealing process of subjecting the cold rolled steel sheet obtained in the second cold rolling process to final annealing under conditions such that a temperature range to be retained is 900° C. or higher and 1200° C. or lower.

[0183] Each process is explained below.

(First Cold Rolling Process)

[0184] In the first cold rolling process, the hot rolled steel sheet having the above chemical composition is subjected to cold rolling at the rolling reduction (cumulative rolling reduction) of 10% or larger and 75% or smaller.

[0185] When the rolling reduction in the first cold rolling process is smaller than 10% or larger than 75%, the intended magnetic characteristics and roundness may not be obtained. Thus, the rolling reduction in the first cold rolling process is to be 10% or larger and 75% or smaller.

[0186] Conditions of the cold rolling other than the above, such as a steel sheet temperature during cold rolling and a diameter of the rolling roll, are not particularly limited, and are appropriately selected depending on the chemical composition of the hot rolled steel sheet, the intended sheet thickness of the steel sheet, or the like.

[0187] In general, the hot rolled steel sheet is subject to cold rolling after a scale formed on a surface of the steel sheet during hot rolling is removed by pickling. As described later, in a case where the hot rolled steel sheet is subjected to hot-band annealing, the hot rolled steel sheet may be pickled either before the hot-band annealing or after the hot-band annealing.

(Intermediate Annealing Process)

[0188] In the intermediate annealing process, the cold rolled steel sheet obtained in the above first cold rolling process is subjected to intermediate annealing under condition such that an average heating rate from 500° C. to 650° C. is 30° C./hour or faster and 1000° C./second or slower (0.0083° C./second or faster and 1000° C./second or slower), a retention temperature is 700° C. or higher and 1100° C. or lower, and a retention time is 10 seconds or longer and 40 hours or shorter (0.0028 hours or longer and 40 hours or shorter).

[0189] When the above conditions are not satisfied in the intermediate annealing process, the intended magnetic characteristics and roundness may not be obtained. Conditions of the intermediate annealing other than the above are not particularly limited.

[0190] The average heating rate from 500° C. to 650° C. is preferably 300° C./second or faster. The retention temperature is preferably 850° C. or higher. The retention time is preferably 180 seconds or shorter (0.05 hours or shorter). In particular, when the Si content of more than 3.25%, the average heating rate from 500° C. to 650° C. of 300° C./second or faster, the retention temperature of 850° C. or higher, and the retention time of 180 seconds or shorter are simultaneously satisfied in addition to satisfying the conditions of the present embodiment, it is possible to obtain the non oriented electrical steel sheet in which both desirable magnetic characteristics and roundness are simultaneously and preferably achieved.

(Second Cold Rolling Process)

[0191] In the second cold rolling process, the intermediate-annealed steel sheet obtained in the above intermediate annealing process is subjected to cold rolling at the rolling reduction (cumulative rolling reduction) of 50% or larger and 85% or smaller to obtain the sheet thickness of 0.10 mm or more and 0.35 mm or less.

[0192] When the rolling reduction in the second cold rolling process is smaller than 50% or larger than 85%, the intended magnetic characteristics and roundness may not be obtained. Thus, the rolling reduction in the second cold rolling process is to be 50% or larger and 85% or smaller.

[0193] The sheet thickness is to be 0.10 mm or more and 0.35 mm or less. The sheet thickness is preferably 0.15 mm or more and 0.30 mm or less.

[0194] Conditions of the cold rolling other than the above, such as a steel sheet temperature during cold rolling and a diameter of the rolling roll, are not particularly limited, and are appropriately selected depending on the chemical composition of the steel sheet, the intended sheet thickness of the steel sheet, or the like.

(Final Annealing Process)

[0195] In the final annealing process, the cold rolled steel sheet obtained in the above second cold rolling process is subjected to final annealing at the temperature range to be retained of 900° C. or higher and 1200° C. or lower.

[0196] When the final annealing temperature in the final annealing process is lower than 900° C., the average grain size may become less than 30 μm due to insufficient grain growth, and thereby sufficient magnetic characteristics may not be obtained. Thus, the final annealing temperature is to be 900° C. or higher. On the other hand, when the final annealing temperature is higher than 1200° C., the grain growth may proceed excessively, the average grain size may become more than 200 μm, and thereby sufficient magnetic characteristics may not be obtained. Thus, the final annealing temperature is to be 1200° C. or lower.

[0197] The final annealing time for retaining the cold rolled steel sheet in the temperature range of 900° C. or higher and 1200° C. or lower may not be particularly specified, but it is preferably 1 second or longer to more reliably obtain favorable magnetic characteristics. On the other hand, from a productive standpoint, the final annealing time is preferably 120 seconds or shorter.

[0198] Conditions of the final annealing other than the above are not particularly limited.

(Hot-Band Annealing Process)

[0199] The hot rolled steel sheet to be subjected to the above first cold rolling process may be subjected to hot-band annealing. When the hot rolled steel sheet is subjected to hot-band annealing, it is possible to obtain favorable magnetic characteristics.

[0200] The hot-band annealing may be performed by either box annealing or continuous annealing. When the box annealing is performed, the hot rolled steel sheet is preferably retained in a temperature range of 700° C. or higher and 900° C. or lower for 1 hour or longer and 20 hours or shorter. When the continuous annealing is performed, the hot rolled steel sheet is preferably retained in a temperature range of 850° C. or higher and 1100° C. or lower for 1 second or longer and 180 seconds or shorter.

[0201] Conditions of the hot-band annealing other than the above are not particularly limited.

(Hot Rolling Process)

[0202] The hot rolled steel sheet to be subjected to the first cold rolling process can be obtained by subjecting a steel ingot or steel piece (hereinafter referred to as "slab") having the above chemical composition to hot rolling.

[0203] In the hot rolling, a steel having the above chemical composition is made into the slab by typical methods such as continuous casting or blooming the steel ingot. The slab is put into a heating furnace and then subjected to hot rolling. At this time, when the slab temperature is high, the hot rolling may be performed without putting the slab into the heating furnace.

[0204] Conditions of the hot-band annealing are not particularly limited.

(Other Processes)

[0205] After the final annealing process, a coating process of applying an insulation coating including only an organic component, only an inorganic component, or an organic-inorganic compound to a surface of the steel sheet may be performed by typical methods. From a standpoint of reducing an environmental load, an insulation coating that does not include chromium may be applied. Moreover, the coating process may be a process of applying an insulation coating that is adhesiveness by heating and pressurizing. As coating material exhibiting adhesiveness, an acrylic resin, a phenol resin, an epoxy resin, a melamine resin, or the like can be used.

(Manufacturing Method of Iron Core and Manufacturing Method of Motor)

[0206] An integrally punched iron core may be manufactured using the non oriented electrical steel sheet according to the present embodiment obtained as described above. A manufacturing method of the iron core may include a process of punching and laminating the above non oriented electrical steel sheet. Moreover, a motor may be manufactured using the integrally punched iron core. A manufacturing method of the motor may include a process of preparing an iron core by punching and laminating the above non oriented electrical steel sheet, and a process of assembling the motor using the iron core.

Examples

[0207] The effects of an aspect of the present invention are described in detail with reference to the following examples. However, the condition in the examples is an example condition employed to confirm the operability and the effects of the present invention, so that the present invention is not limited to the example condition. The present invention can employ various types of conditions as long as the conditions do not depart from the scope of the present invention and can achieve the object of the present invention. Hereinafter, the present invention is explained in detail with reference to examples and comparative examples.

[0208] Non oriented electrical steel sheets were prepared by performing each process under the conditions shown in Tables 1 to 16 using slabs whose chemical compositions were adjusted. Moreover, in a case where hot-band annealing was not performed, pickling was performed after hot rolling. In a case where hot-band annealing was performed, pickling for Test No. 1 and 17 was performed before the hot-band annealing, and pickling for the others was performed after the hot-band annealing. Moreover, a retention time of the final annealing was 30 seconds.

[0209] A chemical composition, a sheet thickness, an average grain size, an X value related to a magnetic flux density, an iron loss $W_{10/1k}$, and a roundness of the prepared non oriented electrical steel sheet were measured. Measurement methods thereof are as described above. Measurement results are shown in Tables 1 to 16. Herein, a chemical composition of the prepared non oriented electrical steel sheet was substantially the same as a chemical composition of the slab. The element represented by "-" in the table indicates that it was not consciously controlled and prepared. Moreover, the Si content indicated by "3.3" in the table was more than 3.25%. Moreover, the manufacturing condition represented by "-" in the table indicates that it was not controlled. Moreover, a sheet thickness of the prepared non oriented electrical steel sheet was the same as a final sheet thickness after second cold rolling process.

[0210] Moreover, as the mechanical anisotropy, a roundness was defined as a ratio of maximum and minimum values of a diameter of an inner circumference of the above ring-shaped core, and the roundness was evaluated using the following criteria.

[0211] Excellent: Roundness is 0.9999 or more and 1.0000 or less.

[0212] Very Good: Roundness is more than 0.9998 and less than 0.9999.

[0213] Good: Roundness is more than 0.9997 and 0.9998 or less.

[0214] Poor: Roundness is 0.9997 or less.

[0215] As shown in Tables 1 to 16, among Test Nos. 1 to 86, inventive examples were excellent in the magnetic characteristics and the roundness as the non oriented electrical steel sheet. On the other hand, among Test Nos. 1 to 86, comparative examples were not excellent in at least one of the magnetic characteristics and the roundness.

TABLE 1

		MANUFACTURING RESULTS CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)														
No.	STEEL TYPE	C	Si	Mn	sol. Al	P	S	N	B	0	Mg	Ca	Ti	V	Cr	Ni
1	S1	0.002	1.8	0.2	2.3	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
2	S1	0.002	1.8	0.2	2.3	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
3	S1	0.002	1.8	0.2	2.3	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
4	S1	0.002	1.8	0.2	2.3	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
5	S1	0.002	1.8	0.2	2.3	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
6	S1	0.002	1.8	0.2	2.3	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
7	S1	0.002	1.8	0.2	2.3	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
8	S1	0.002	1.8	0.2	2.3	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
9	S2	0.002	2.0	2.0	2.0	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
10	S3	0.002	2.9	0.2	1.1	0.01	0.001	0.002	—	0.002	—	—	—	—	—	—
11	S4	0.002	2.0	0.2	1.0	0.08	0.003	0.002	—	0.002	—	—	—	—	—	—
12	S5	0.002	2.3	1.2	1.7	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
13	S6	0.002	2.3	1.2	1.7	0.01	0.003	0.002	—	0.002	—	0.003	—	—	—	—
14	S7	0.002	2.3	1.2	1.7	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
15	S8	0.002	2.3	1.2	1.7	0.01	0.003	0.002	—	0.002	—	0.003	—	—	—	—
16	S2	0.002	2.0	2.0	2.0	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
17	S2	0.002	2.0	2.0	2.0	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
18	S2	0.002	2.0	2.0	2.0	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
19	S2	0.002	2.0	2.0	2.0	0.01	0.003	0.002	—	0.002	—	—	—	—	—	—
20	S9	0.002	3.3	1.2	1.5	0.02	0.001	0.002	—	0.002	—	—	—	—	—	—
21	S10	0.002	3.3	0.9	0.7	0.01	0.001	0.002	—	0.002	—	—	—	—	—	—
22	S11	0.002	3.4	0.9	1.5	0.02	0.001	0.002	—	0.002	—	—	—	—	—	—
23	S9	0.002	3.3	1.2	1.5	0.02	0.001	0.002	—	0.002	—	—	—	—	—	—
24	S12	0.002	3.6	0.9	0.7	0.01	0.001	0.002	—	0.002	—	—	—	—	—	—
25	S13	0.002	3.7	0.5	1.1	0.03	0.002	0.002	—	0.002	—	—	—	—	—	—

TABLE 2

		MANUFACTURING RESULTS CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)														
No.	STEEL TYPE	C	Si	Mn	sol. Al	P	S	N	B	O	Mg	Ca	Ti	V	Cr	Ni
26	S14	0.003	3.5	0.5	0.3	0.03	0.002	0.002	—	0.002	—	—	—	—	—	—
27	S15	0.002	3.6	0.5	0.3	0.03	0.001	0.002	—	0.002	—	—	—	—	—	—
28	S16	0.003	3.7	0.5	0.3	0.03	0.002	0.002	—	0.002	—	—	—	—	—	—
29	S17	0.002	3.4	1.2	1.7	0.05	0.002	0.002	—	0.002	—	—	—	—	—	—
30	S18	0.002	3.6	1.2	1.7	0.05	0.002	0.002	—	0.002	—	—	—	—	—	—
31	S19	0.002	3.8	1.2	1.7	0.20	0.002	0.002	—	0.002	—	—	—	—	—	—

TABLE 2-continued

		MANUFACTURING RESULTS CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)														
No.	STEEL TYPE	C	Si	Mn	sol. Al	P	S	N	B	O	Mg	Ca	Ti	V	Cr	Ni
32	S20	0.002	3.3	0.9	0.7	0.01	0.001	0.002	0.0002	0.002	—	—	—	—	—	—
33	S21	0.002	3.4	0.9	1.5	0.02	0.001	0.002	—	0.002	0.0001	—	—	—	—	—
34	S22	0.002	3.3	1.2	1.5	0.02	0.001	0.002	—	0.002	—	—	0.0015	—	—	—
35	S23	0.002	3.6	0.9	0.7	0.01	0.001	0.002	—	0.002	—	—	—	0.0015	—	—
36	S24	0.002	3.7	0.5	1.1	0.03	0.002	0.002	—	0.002	—	—	—	—	0.002	—
37	S25	0.003	3.5	0.5	0.3	0.03	0.002	0.002	—	0.002	—	—	—	—	—	0.002
38	S26	0.002	3.6	0.5	0.3	0.03	0.001	0.002	—	0.002	—	—	—	—	—	—
39	S27	0.003	3.7	0.5	0.3	0.03	0.002	0.002	—	0.002	—	—	—	—	—	—
40	S28	0.002	3.4	1.2	1.7	0.05	0.002	0.002	—	0.002	—	0.003	—	—	—	—
41	S29	0.002	3.5	1.2	1.7	0.05	0.002	0.002	—	0.002	—	—	—	—	—	—
42	S30	0.002	3.7	1.1	1.3	0.05	0.002	0.002	—	0.002	—	—	—	—	—	—
43	S31	0.002	3.3	0.9	0.7	0.01	0.001	0.002	0.0003	0.002	—	—	—	—	—	—
44	S32	0.002	3.4	0.9	1.3	0.02	0.001	0.002	—	0.002	0.0005	—	—	—	—	—
45	S33	0.001	3.3	1.2	1.0	0.02	0.001	0.002	—	0.002	—	—	0.0018	—	—	—
46	S34	0.001	3.6	0.9	0.7	0.01	0.001	0.002	—	0.002	—	—	—	0.0018	—	—
47	S35	0.001	3.7	0.5	1.1	0.03	0.002	0.002	—	0.002	—	—	—	—	0.035	—
48	S36	0.003	3.5	0.5	0.3	0.03	0.002	0.002	—	0.002	—	—	—	—	—	0.041
49	S37	0.002	3.6	0.5	0.3	0.03	0.001	0.002	—	0.002	—	—	—	—	—	—
50	S38	0.003	3.7	0.2	0.3	0.03	0.002	0.002	—	0.002	—	—	—	—	—	—

TABLE 3

		MANUFACTURING RESULTS CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)														
No.	STEEL TYPE	C	Si	Mn	sol. Al	P	S	N	B	O	Mg	Ca	Ti	V	Cr	Ni
51	S39	0.002	3.4	1.2	1.2	0.05	0.002	0.002	—	0.002	—	—	—	—	—	—
52	S40	0.002	3.6	1.2	1.1	0.05	0.002	0.002	—	0.002	—	—	—	—	—	—
53	S41	0.002	3.8	1.2	0.9	0.05	0.002	0.002	—	0.002	—	—	—	—	—	—
54	S42	0.002	3.3	0.9	0.7	0.01	0.001	0.002	0.0030	0.002	—	—	—	—	—	—
55	S43	0.002	3.4	0.9	0.8	0.02	0.001	0.002	—	0.002	0.0150	—	—	—	—	—
56	S44	0.001	3.3	1.2	0.8	0.02	0.001	0.002	—	0.002	—	—	0.0025	—	—	—
57	S45	0.001	3.6	0.9	0.7	0.01	0.001	0.002	—	0.002	—	—	—	0.0035	—	—
58	S46	0.001	3.7	0.5	0.3	0.03	0.002	0.002	—	0.002	—	—	—	—	1.500	—
59	S47	0.003	3.5	0.5	0.3	0.03	0.002	0.002	—	0.002	—	—	—	—	—	1.200
60	S48	0.002	3.6	0.2	0.3	0.03	0.001	0.002	—	0.002	—	—	—	—	—	—
61	S49	0.003	3.7	0.5	0.3	0.03	0.002	0.002	—	0.002	—	—	—	—	—	—
62	S50	0.002	3.4	1.2	0.8	0.05	0.002	0.002	—	0.002	—	—	—	—	—	—
63	S51	0.002	3.8	0.2	0.3	0.01	0.002	0.002	—	0.002	—	—	—	—	—	—
64	S52	0.002	4.2	0.2	0.3	0.01	0.002	0.002	—	0.002	—	—	—	—	—	—
65	S53	0.002	0.8	0.2	1.2	0.01	0.001	0.002	0.0003	0.002	—	—	—	—	—	—
66	S54	0.002	5.5	1.2	1.7	0.05	0.002	0.002	—	0.002	—	—	—	—	—	—
67	S55	0.002	1.2	3.5	0.3	0.02	0.001	0.002	—	0.002	0.0005	—	—	—	—	—
68	S56	0.001	1.5	1.2	3.5	0.02	0.001	0.002	—	0.002	—	—	—	—	—	—
69	S57	0.001	1.5	0.9	0.7	0.35	0.001	0.002	—	0.002	—	—	—	—	—	—
70	S58	0.001	1.2	0.5	1.1	0.03	0.020	0.002	—	0.002	—	—	—	—	—	—
71	S59	0.003	2.0	0.5	0.3	0.03	0.002	0.020	—	0.002	—	—	0.085	—	—	—
72	S60	0.001	2.8	0.5	1.1	0.03	0.001	0.002	—	0.002	—	—	—	—	5.600	—
73	S61	0.003	2.8	0.5	0.3	0.03	0.002	0.002	—	0.002	—	—	—	—	—	5.200
74	S62	0.002	2.8	0.5	0.3	0.03	0.001	0.002	—	0.002	—	—	—	—	—	—
75	S63	0.003	1.2	0.5	0.3	0.03	0.009	0.008	0.110	0.080	0.150	0.020	—	—	—	—

TABLE 4

		MANUFACTURING RESULTS CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)														
No.	STEEL TYPE	C	Si	Mn	sol. Al	P	S	N	B	O	Mg	Ca	Ti	V	Cr	Ni
76	S64	0.002	3.9	0.2	2.1	0.15	0.002	0.002	—	0.002	—	—	—	—	—	—
77	S65	0.002	3.9	0.2	2.1	0.15	0.009	0.002	—	0.002	—	—	—	—	—	—
78	S66	0.002	1.2	0.2	2.1	0.06	0.002	0.002	—	0.002	—	—	0.153	0.123	—	—
79	S67	0.002	1.2	0.2	2.1	0.06	0.002	0.002	—	0.002	—	—	—	—	—	—

TABLE 4-continued

		MANUFACTURING RESULTS CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)														
No.	STEEL TYPE	C	Si	Mn	sol. Al	P	S	N	B	O	Mg	Ca	Ti	V	Cr	Ni
80	S68	0.002	1.2	0.2	2.1	0.06	0.002	0.002	—	0.002	—	—	—	—	—	—
81	S69	0.002	2.5	0.9	1.1	0.02	0.001	0.002	—	0.002	0.0001	—	—	—	—	—
82	S69	0.002	2.5	0.9	1.1	0.02	0.001	0.002	—	0.002	0.0001	—	—	—	—	—
83	S69	0.002	2.5	0.9	1.1	0.02	0.001	0.002	—	0.002	0.0001	—	—	—	—	—
84	S70	0.002	3.3	0.9	1.1	0.02	0.001	0.002	—	0.002	0.0001	—	—	—	—	—
85	S70	0.002	3.3	0.9	1.1	0.02	0.001	0.002	—	0.002	0.0001	—	—	—	—	—
86	S71	0.002	2.8	0.9	2.3	0.02	0.001	0.002	—	0.002	0.0001	—	—	—	—	—

TABLE 5

		MANUFACTURING RESULTS													
		CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)													
No.	STEEL TYPE	Cu	Zr	Sn	Sb	Ce	Nd	Bi	W	Mo	Nb	Y	Si + sol. Al	DENSITY g/cm^3	Is T
1	S1	—	—	—	—	—	—	—	—	—	—	—	4.1	7.516	1.973
2	S1	—	—	—	—	—	—	—	—	—	—	—	4.1	7.516	1.973
3	S1	—	—	—	—	—	—	—	—	—	—	—	4.1	7.516	1.973
4	S1	—	—	—	—	—	—	—	—	—	—	—	4.1	7.516	1.973
5	S1	—	—	—	—	—	—	—	—	—	—	—	4.1	7.516	1.973
6	S1	—	—	—	—	—	—	—	—	—	—	—	4.1	7.516	1.973
7	S1	—	—	—	—	—	—	—	—	—	—	—	4.1	7.516	1.973
8	S1	—	—	—	—	—	—	—	—	—	—	—	4.1	7.516	1.973
9	S2	—	—	—	—	—	—	—	—	—	—	—	4.0	7.520	1.939
10	S3	—	—	—	—	—	—	—	—	—	—	—	4.0	7.570	1.989
11	S4	—	—	—	—	—	—	—	—	—	—	—	3.0	7.640	2.027
12	S5	—	—	0.020	—	—	—	—	—	—	—	—	4.0	7.554	1.964
13	S6	—	—	0.020	—	—	—	—	—	—	—	—	4.0	7.546	1.962
14	S7	—	—	—	0.030	—	—	—	—	—	—	—	4.0	7.542	1.961
15	S8	—	—	—	0.030	—	—	—	—	—	—	—	4.0	7.532	1.958
16	S2	—	—	—	—	—	—	—	—	—	—	—	4.0	7.520	1.939
17	S2	—	—	—	—	—	—	—	—	—	—	—	4.0	7.520	1.939
18	S2	—	—	—	—	—	—	—	—	—	—	—	4.0	7.520	1.939
19	S2	—	—	—	—	—	—	—	—	—	—	—	4.0	7.520	1.939
20	S9	—	—	—	—	—	—	—	—	—	—	—	4.8	7.480	1.928
21	S10	—	—	—	—	—	—	—	—	—	—	—	4.0	7.569	1.974
22	S11	—	—	—	—	—	—	—	—	—	—	—	4.9	7.476	1.932
23	S9	—	—	—	—	—	—	—	—	—	—	—	4.8	7.480	1.928
24	S12	—	—	—	—	—	—	—	—	—	—	—	4.3	7.549	1.963
25	S13	—	—	—	—	—	—	—	—	—	—	—	4.8	7.503	1.949

TABLE 6

		MANUFACTURING RESULTS													
		CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)													
No.	STEEL TYPE	Cu	Zr	Sn	Sb	Ce	Nd	Bi	W	Mo	Nb	Y	Si + sol. Al	DENSITY g/cm^3	Is T
26	S14	—	—	—	—	—	—	—	—	—	—	—	3.8	7.602	1.995
27	S15	—	—	—	—	—	—	—	—	—	—	—	3.9	7.596	1.992
28	S16	—	—	—	—	—	—	—	—	—	—	—	4.0	7.590	1.988
29	S17	—	—	—	—	—	—	—	—	—	—	—	5.1	7.452	1.914
30	S18	—	—	—	—	—	—	—	—	—	—	—	5.3	7.439	1.907
31	S19	—	—	—	—	—	—	—	—	—	—	—	5.5	7.426	1.897
32	S20	—	—	—	—	—	0.003	—	—	—	—	—	4.0	7.569	1.974
33	S21	—	—	—	—	—	—	0.002	—	—	—	—	4.9	7.476	1.931
34	S22	—	—	—	—	—	—	—	0.003	—	—	—	4.8	7.480	1.928
35	S23	—	—	—	—	—	—	—	—	0.003	—	—	4.3	7.549	1.963
36	S24	—	—	—	—	—	—	—	—	—	0.002	—	4.8	7.503	1.949
37	S25	—	—	—	—	—	—	—	—	—	—	0.003	3.8	7.602	1.995
38	S26	0.002	—	—	—	—	—	—	—	—	—	—	3.9	7.596	1.991

TABLE 6-continued

		MANUFACTURING RESULTS													
		CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)													
No.	STEEL TYPE	Cu	Zr	Sn	Sb	Ce	Nd	Bi	W	Mo	Nb	Y	Si + sol. Al	DENSITY g/cm^3	Is T
39	S27	—	0.0005	—	—	—	—	—	—	—	—	—	4.0	7.590	1.988
40	S28	—	—	0.031	—	—	—	—	—	—	—	—	5.1	7.452	1.914
41	S29	—	—	—	0.005	—	—	—	—	—	—	—	5.2	7.445	1.911
42	S30	—	—	—	—	0.003	—	—	—	—	—	—	5.0	7.477	1.925
43	S31	—	—	—	—	—	0.005	—	—	—	—	—	4.0	7.569	1.974
44	S32	—	—	—	—	—	—	0.003	—	—	—	—	4.7	7.497	1.941
45	S33	—	—	—	—	—	—	—	0.005	—	—	—	4.3	7.534	1.953
46	S34	—	—	—	—	—	—	—	—	0.005	—	—	4.3	7.549	1.963
47	S35	—	—	—	—	—	—	—	—	—	0.002	—	4.8	7.503	1.948
48	S36	—	—	—	—	—	—	—	—	—	—	0.004	3.8	7.602	1.994
49	S37	0.047	—	—	—	—	—	—	—	—	—	—	3.9	7.596	1.991
50	S38	—	0.0081	—	—	—	—	—	—	—	—	—	4.0	7.592	1.994

TABLE 7

		MANUFACTURING RESULTS													
		CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)													
No.	STEEL TYPE	Cu	Zr	Sn	Sb	Ce	Nd	Bi	W	Mo	Nb	Y	Si + sol. Al	DENSITY g/cm^3	Is T
51	S39	—	—	0.055	—	—	—	—	—	—	—	—	4.6	7.506	1.937
52	S40	—	—	—	0.048	—	—	—	—	—	—	—	4.7	7.504	1.935
53	S41	—	—	—	—	0.004	—	—	—	—	—	—	4.7	7.512	1.938
54	S42	—	—	—	—	—	0.005	—	—	—	—	—	4.0	7.569	1.974
55	S43	—	—	—	—	—	—	0.003	—	—	—	—	4.2	7.551	1.965
56	S44	—	—	—	—	—	—	—	0.009	—	—	—	4.1	7.555	1.962
57	S45	—	—	—	—	—	—	—	—	0.009	—	—	4.3	7.549	1.963
58	S46	—	—	—	—	—	—	—	—	—	0.020	—	4.0	7.590	1.956
59	S47	—	—	—	—	—	—	—	—	—	—	0.040	3.8	7.602	1.969
60	S48	1.800	—	—	—	—	—	—	—	—	—	—	3.9	7.598	1.961
61	S49	—	0.0081	—	—	—	—	—	—	—	—	—	4.0	7.590	1.988
62	S50	—	—	0.080	—	—	—	—	—	—	—	—	4.2	7.549	1.956
63	S51	—	—	—	0.090	—	—	—	—	—	—	—	4.1	7.586	1.989
64	S52	—	—	—	—	0.040	—	—	—	—	—	—	4.5	7.560	1.975
65	S53	—	—	—	—	—	0.080	—	—	—	—	—	2.0	7.681	2.059
66	S54	—	—	—	—	—	—	—	—	—	—	—	7.2	7.317	1.838
67	S55	—	—	—	—	—	—	—	—	—	—	—	1.5	7.725	2.013
68	S56	—	—	—	—	—	—	—	—	—	—	—	5.0	7.379	1.898
69	S57	—	—	—	—	—	—	—	—	—	—	—	2.2	7.683	2.035
70	S58	—	—	—	—	—	—	—	—	—	—	—	2.3	7.663	2.042
71	S59	—	—	—	—	—	—	—	—	—	—	—	2.3	7.698	2.050
72	S60	—	—	—	—	—	—	—	—	—	—	—	3.9	7.561	1.866
73	S61	—	—	—	—	—	—	—	—	—	—	—	3.1	7.647	1.913
74	S62	5.300	—	—	—	—	—	—	—	—	—	—	3.1	7.647	1.911
75	S63	—	0.153	—	—	—	—	—	—	—	—	—	1.5	7.750	2.072

TABLE 8

		MANUFACTURING RESULTS													
		CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)													
No.	STEEL TYPE	Cu	Zr	Sn	Sb	Ce	Nd	Bi	W	Mo	Nb	Y	Si + sol. Al	DENSITY g/cm^3	Is T
76	S64	—	—	0.210	—	—	—	—	—	—	—	—	6.0	7.384	1.893
77	S65	—	—	—	0.210	0.150	—	—	—	—	—	—	6.0	7.384	1.890
78	S66	—	—	—	—	—	—	—	—	—	0.159	—	3.3	7.558	1.990
79	S67	—	—	—	—	—	0.130	0.142	—	—	—	0.151	3.3	7.558	1.991
80	S68	—	—	—	—	—	—	—	0.152	0.154	—	—	3.3	7.558	1.993
81	S69	—	—	—	—	—	—	—	—	—	—	—	3.6	7.577	1.985
82	S69	—	—	—	—	—	—	—	—	—	—	—	3.6	7.577	1.985
83	S69	—	—	—	—	—	—	—	—	—	—	—	3.6	7.577	1.985

TABLE 8-continued

		MANUFACTURING RESULTS													
		CHEMICAL COMPOSITION (UNIT: mass %, BALANCE CONSISTING OF Fe AND IMPURITIES)													
No.	STEEL TYPE	Cu	Zr	Sn	Sb	Ce	Nd	Bi	W	Mo	Nb	Y	Si + sol. Al	DENSITY g/cm^3	Is T
84	S70	—	—	—	—	—	—	—	—	—	—	—	4.4	7.525	1.955
85	S70	—	—	—	—	—	—	—	—	—	—	—	4.4	7.525	1.955
86	S71	—	—	—	—	—	—	—	—	—	—	—	5.1	7.428	1.915

TABLE 9

		MANUFACTURING CONDITIONS										
			HOT BAND ANNEALING		FIRST COLD ROLLING		INTERMEDIATE ANNEALING			SECOND COLD ROLLING		
No.	STEEL TYPE	SHEET THICKNESS OF HOT ROLLED STEEL SHEET mm	RETENTION TEMPERATURE ° C.	RETENTION TIME hour	ROLLING REDUCTION %	INTERMEDIATE SHEET THICKNESS mm	RETENTION TEMPERATURE ° C.	RETENTION TIME hour	AVERAGE HEATING RATE FROM 500° C. TO 650° C. ° C./sec	ROLLING REDUCTION %	FINAL SHEET THICKNESS mm	FINAL ANNEALING TEMPERATURE ° C.
1	S1	2.20	900	10	31.8	1.50	800	10	0.011	80.0	0.30	1050
2	S1	2.30	—	—	65.2	0.80	800	10	0.011	75.0	0.20	1180
3	S1	2.30	—	—	73.9	0.60	800	10	0.011	66.8	0.20	1080
4	S1	2.20	—	—	68.2	0.70	750	30	0.011	64.3	0.25	950
5	S1	2.20	—	—	60.9	0.86	900	10	0.011	65.1	0.30	1050
6	S1	2.30	—	—	56.5	1.00	800	10	0.011	59.9	0.40	1000
7	S1	2.19	—	—	56.5	0.95	800	10	0.011	47.4	0.50	950
8	S1	2.16	—	—	65.2	0.75	800	10	0.011	46.7	0.40	1140
9	S2	2.00	—	—	55.0	0.90	750	10	0.011	72.2	0.25	1050
10	S3	2.00	—	—	57.0	0.86	800	10	0.011	65.1	0.30	1100
11	S4	2.00	—	—	55.0	0.90	800	10	0.011	66.7	0.30	1140
12	S5	2.00	—	—	20.0	1.60	800	4	0.011	81.3	0.30	1100
13	S6	2.00	—	—	20.0	1.60	800	4	0.011	81.3	0.30	1100
14	S7	2.00	—	—	20.0	1.60	800	4	0.011	81.3	0.30	1100
15	S8	2.00	—	—	20.0	1.60	800	4	0.011	81.3	0.30	1100
16	S2	2.00	—	—	55.0	0.90	800	10	0.011	72.2	0.25	1100
17	S2	2.00	800	10	55.0	0.90	800	10	0.011	72.2	0.25	1100
18	S2	2.00	950	0.0056	55.0	0.90	800	10	0.011	72.2	0.25	1100
19	S2	2.30	1000	0.0167	70.0	0.69	1000	0.017	400	64.0	0.25	1000
20	S9	2.00	950	0.0056	55.0	0.90	800	10	0.011	72.2	0.25	1100
21	S10	2.30	1000	0.0167	70.0	0.69	1000	0.017	350	64.0	0.25	1000
22	S11	2.30	1000	0.0167	73.0	0.62	1000	0.017	350	60.0	0.25	1000
23	S9	2.30	1000	0.0167	70.0	0.69	1000	0.017	400	64.0	0.25	1000
24	S12	2.30	1000	0.0167	70.0	0.69	1000	0.017	500	64.0	0.25	1000
25	S13	2.00	1000	0.0167	70.0	0.60	1000	0.017	600	59.0	0.25	1000

TABLE 10

No.	STEEL TYPE	MANUFACTURING CONDITIONS: SHEET THICKNESS OF HOT ROLLED STEEL SHEET mm	HOT BAND ANNEALING: RETENTION TEMPERATURE ° C.	HOT BAND ANNEALING: RETENTION TIME hour	FIRST COLD ROLLING: ROLLING REDUCTION %	FIRST COLD ROLLING: INTERMEDIATE SHEET THICKNESS mm	INTERMEDIATE ANNEALING: RETENTION TEMPERATURE ° C.	INTERMEDIATE ANNEALING: RETENTION TIME hour	INTERMEDIATE ANNEALING: AVERAGE HEATING RATE FROM 500° C. TO 650° C. ° C./sec	SECOND COLD ROLLING: ROLLING REDUCTION %	SECOND COLD ROLLING: FINAL SHEET THICKNESS mm	FINAL ANNEALING TEMPERATURE ° C.
26	S14	2.00	1000	0.0167	60.0	0.80	1000	0.017	700	69.0	0.25	1000
27	S15	2.00	980	0.0167	70.0	0.60	1000	0.017	800	59.0	0.25	1000
28	S16	2.00	1000	0.0167	70.0	0.60	1000	0.017	900	59.0	0.25	1000
29	S17	2.00	1000	0.0167	70.0	0.60	1000	0.017	400	59.0	0.25	1000
30	S18	2.00	1000	0.0167	70.0	0.60	1000	0.017	500	59.0	0.25	1000
31	S19	2.00	1050	0.0167	70.0	0.60	1000	0.017	600	59.0	0.25	1000
32	S20	2.00	1050	0.0333	30.0	1.40	1050	0.008	350	82.0	0.25	980
33	S21	2.00	1000	0.0333	30.0	1.40	1050	0.008	350	82.0	0.25	980
34	S22	2.00	980	0.0333	30.0	1.40	1050	0.008	400	82.0	0.25	980
35	S23	2.00	950	0.0333	30.0	1.40	1050	0.008	500	82.0	0.25	980
36	S24	2.00	1000	0.0333	30.0	1.40	1050	0.008	600	82.0	0.25	950
37	S25	2.00	1050	0.0333	30.0	1.40	1050	0.008	700	82.0	0.25	980
38	S26	2.00	1000	0.0333	30.0	1.40	1050	0.008	800	82.0	0.25	980
39	S27	2.00	1000	0.0333	30.0	1.40	1050	0.008	900	82.0	0.25	980
40	S28	2.00	1050	0.0333	30.0	1.40	1050	0.008	400	82.0	0.25	980
41	S29	2.00	950	0.0333	30.0	1.40	1050	0.008	500	82.0	0.25	980
42	S30	2.00	950	0.0333	30.0	1.40	1050	0.008	600	82.0	0.25	1000
43	S31	1.90	980	0.0278	12.0	1.67	1050	0.017	350	85.0	0.25	980
44	S32	1.90	980	0.0278	12.0	1.67	1050	0.017	350	85.0	0.25	1000
45	S33	1.90	980	0.0278	12.0	1.67	1050	0.017	400	85.0	0.25	980
46	S34	1.90	980	0.0278	12.0	1.67	1050	0.017	500	85.0	0.25	1000
47	S35	1.90	1050	0.0278	12.0	1.67	1050	0.017	600	85.0	0.25	980
48	S36	1.90	1000	0.0278	12.0	1.67	1050	0.017	700	85.0	0.25	1000
49	S37	1.90	980	0.0278	12.0	1.67	1050	0.017	800	85.0	0.25	980
50	S38	1.90	1050	0.0278	12.0	1.67	1050	0.017	900	85.0	0.25	1000

TABLE 11

No.	STEEL TYPE	MANUFACTURING CONDITIONS: SHEET THICKNESS OF HOT ROLLED STEEL SHEET mm	HOT BAND ANNEALING: RETENTION TEMPERATURE ° C.	HOT BAND ANNEALING: RETENTION TIME hour	FIRST COLD ROLLING: ROLLING REDUCTION %	FIRST COLD ROLLING: INTERMEDIATE SHEET THICKNESS mm	INTERMEDIATE ANNEALING: RETENTION TEMPERATURE ° C.	INTERMEDIATE ANNEALING: RETENTION TIME hour	INTERMEDIATE ANNEALING: AVERAGE HEATING RATE FROM 500° C. TO 650° C. ° C./sec	SECOND COLD ROLLING: ROLLING REDUCTION %	SECOND COLD ROLLING: FINAL SHEET THICKNESS mm	FINAL ANNEALING TEMPERATURE ° C.
51	S39	1.90	980	0.0278	12.0	1.67	1050	0.017	400	85.0	0.25	980
52	S40	1.90	980	0.0278	12.0	1.67	1050	0.017	500	85.0	0.25	1000
53	S41	1.90	980	0.0278	12.0	1.67	1050	0.017	600	85.0	0.25	1000
54	S42	1.90	850	0.0167	12.0	1.67	850	0.008	350	85.0	0.25	1000
55	S43	1.90	850	0.0167	12.0	1.67	850	0.008	350	85.0	0.25	1000
56	S44	1.90	850	0.0167	12.0	1.67	850	0.008	400	85.0	0.25	1000
57	S45	1.90	850	0.0167	12.0	1.67	850	0.008	500	85.0	0.25	1000
58	S46	1.90	850	0.0167	12.0	1.67	850	0.008	600	85.0	0.25	1000
59	S47	1.90	850	0.0167	12.0	1.67	850	0.008	700	85.0	0.25	1000
60	S48	1.90	850	0.0167	12.0	1.67	850	0.008	800	85.0	0.25	1000
61	S49	1.90	850	0.0167	12.0	1.67	850	0.008	900	85.0	0.25	1000
62	S50	1.90	850	0.0167	12.0	1.67	850	0.008	400	85.0	0.25	1000
63	S51	1.90	850	0.0167	12.0	1.67	850	0.008	500	85.0	0.25	1000
64	S52	1.90	850	0.0167	12.0	1.67	850	0.008	600	85.0	0.25	1000

TABLE 11-continued

		MANUFACTURING CONDITIONS										
			HOT BAND ANNEALING		FIRST COLD ROLLING		INTERMEDIATE ANNEALING			SECOND COLD ROLLING		FINAL
No.	STEEL TYPE	SHEET THICKNESS OF HOT ROLLED STEEL SHEET mm	RETENTION TEMPERATURE ° C.	RETENTION TIME hour	ROLLING REDUCTION %	INTERMEDIATE SHEET THICKNESS mm	RETENTION TEMPERATURE ° C.	RETENTION TIME hour	AVERAGE HEATING RATE FROM 500° C. TO 650° C. ° C./sec	ROLLING REDUCTION %	FINAL SHEET THICKNESS mm	ANNEALING TEMPERATURE ° C.
65	S53	2.00	1000	0.0500	55.0	0.90	1050	0.008	320	72.2	0.25	900
66	S54	2.00	1000	0.0500	FRACTURE	—	—	—	—	—	—	—
67	S55	2.00	1000	0.0500	55.0	0.90	1050	0.008	280	72.2	0.25	1100
68	S56	2.00	1000	0.0500	55.0	0.90	820	0.008	280	61.0	0.35	1100
69	S57	2.00	1000	0.0500	FRACTURE	—	—	—	—	—	—	—
70	S58	2.00	800	0.0500	0.0	2.00	—	—	—	82.5	0.35	900
71	S59	2.00	800	0.0500	0.0	2.00	—	—	—	82.5	0.35	900
72	S60	2.00	1000	0.0500	55.0	0.90	820	0.008	280	61.0	0.35	1000
73	S61	2.00	1000	0.0500	FRACTURE	—	—	—	—	—	—	—
74	S62	2.00	1000	0.0500	55.0	0.90	820	0.008	280	61.0	0.35	900
75	S63	2.00	1000	0.0500	55.0	0.90	820	0.008	150	61.0	0.35	900

TABLE 12

		MANUFACTURING CONDITIONS										
			HOT BAND ANNEALING		FIRST COLD ROLLING		INTERMEDIATE ANNEALING			SECOND COLD ROLLING		FINAL
No.	STEEL TYPE	SHEET THICKNESS OF HOT ROLLED STEEL SHEET mm	RETENTION TEMPERATURE ° C.	RETENTION TIME hour	ROLLING REDUCTION %	INTERMEDIATE SHEET THICKNESS mm	RETENTION TEMPERATURE ° C.	RETENTION TIME hour	AVERAGE HEATING RATE FROM 500° C. TO 650° C. ° C./sec	ROLLING REDUCTION %	FINAL SHEET THICKNESS mm	ANNEALING TEMPERATURE ° C.
76	S64	2.00	1000	0.0500	FRACTURE	—	—	—	—	—	—	—
77	S65	2.00	1000	0.0500	FRACTURE	—	—	—	—	—	—	—
78	S66	2.00	800	0.0500	12.0	1.76	830	0.008	250	80.0	0.35	900
79	S67	2.00	800	0.0500	12.0	1.76	830	0.008	250	80.0	0.35	900
80	S68	2.00	800	0.0500	12.0	1.76	830	0.008	250	80.0	0.35	900
81	S69	2.00	780	0.0500	0.0	2.00	—	—	—	92.5	0.15	900
82	S69	2.00	780	0.0500	20.0	1.60	750	0.008	250	95.0	0.08	1150
83	S69	2.00	780	0.0500	60.0	0.80	750	0.008	250	50.0	0.40	1150
84	S70	2.00	1000	0.0333	49.0	1.02	1050	0.008	500	85.0	0.15	1000
85	S70	2.00	980	0.0333	30.0	1.40	1050	0.008	600	75.0	0.35	1000
86	S71	2.00	700	0.0333	49.0	1.02	800	0.008	1500	85.0	0.15	1000

TABLE 13

No.	STEEL TYPE	MANUFACTURING RESULTS AVERAGE GRAIN SIZE μm	$W_{10/1k}$ W/kg	B_{50} L DIRECTION T	C DIRECTION T	X VALUE	ROUNDNESS	NOTE
1	S1	78	65	1.709	1.621	0.851	Good	INVENTIVE EXAMPLE
2	S1	162	47	1.720	1.609	0.853	Good	INVENTIVE EXAMPLE
3	S1	90	42	1.708	1.593	0.846	Good	INVENTIVE EXAMPLE
4	S1	51	56	1.707	1.590	0.845	Good	INVENTIVE EXAMPLE
5	S1	80	67	1.736	1.610	0.859	Good	INVENTIVE EXAMPLE
6	S1	65	89	1.705	1.598	0.846	Good	COMPARATIVE EXAMPLE
7	S1	55	110	1.684	1.592	0.838	Good	COMPARATIVE EXAMPLE
8	S1	120	91	1.701	1.596	0.844	Good	COMPARATIVE EXAMPLE
9	S2	82	51	1.681	1.579	0.849	Good	INVENTIVE EXAMPLE
10	S3	98	62	1.721	1.607	0.846	Good	INVENTIVE EXAMPLE
11	S4	115	69	1.753	1.643	0.847	Good	INVENTIVE EXAMPLE
12	S5	95	60	1.705	1.624	0.854	Good	INVENTIVE EXAMPLE
13	S6	107	63	1.686	1.618	0.848	Good	INVENTIVE EXAMPLE
14	S7	111	61	1.691	1.623	0.851	Good	INVENTIVE EXAMPLE
15	S8	90	62	1.679	1.616	0.847	Good	INVENTIVE EXAMPLE
16	S2	101	50	1.692	1.582	0.854	Good	INVENTIVE EXAMPLE
17	S2	96	48	1.709	1.599	0.862	Good	INVENTIVE EXAMPLE
18	S2	89	49	1.704	1.593	0.860	Good	INVENTIVE EXAMPLE
19	S2	78	51	1.641	1.603	0.840	Good	INVENTIVE EXAMPLE
20	S9	90	50	1.698	1.586	0.861	Good	INVENTIVE EXAMPLE
21	S10	193	49	1.681	1.621	0.841	Very Good	INVENTIVE EXAMPLE
22	S11	44	48	1.645	1.601	0.844	Very Good	INVENTIVE EXAMPLE
23	S9	78	45	1.642	1.601	0.844	Very Good	INVENTIVE EXAMPLE
24	S12	78	45	1.672	1.604	0.840	Very Good	INVENTIVE EXAMPLE
25	S13	80	44	1.653	1.603	0.840	Very Good	INVENTIVE EXAMPLE

TABLE 14

No.	STEEL TYPE	MANUFACTURING RESULTS AVERAGE GRAIN SIZE μm	$W_{10/1k}$ W/kg	B_{50} L DIRECTION T	C DIRECTION T	X VALUE	ROUNDNESS	NOTE
26	S14	85	45	1.708	1.615	0.840	Very Good	INVENTIVE EXAMPLE
27	S15	79	44	1.708	1.615	0.842	Very Good	INVENTIVE EXAMPLE
28	S16	80	42	1.708	1.615	0.844	Very Good	INVENTIVE EXAMPLE
29	S17	75	41	1.624	1.601	0.844	Very Good	INVENTIVE EXAMPLE
30	S18	79	43	1.615	1.601	0.844	Very Good	INVENTIVE EXAMPLE
31	S19	80	43	1.602	1.600	0.844	Very Good	INVENTIVE EXAMPLE
32	S20	79	44	1.660	1.610	0.832	Excellent	INVENTIVE EXAMPLE
33	S21	80	45	1.620	1.605	0.836	Excellent	INVENTIVE EXAMPLE
34	S22	83	43	1.620	1.605	0.838	Excellent	INVENTIVE EXAMPLE
35	S23	85	44	1.635	1.621	0.830	Excellent	INVENTIVE EXAMPLE
36	S24	79	45	1.633	1.606	0.833	Excellent	INVENTIVE EXAMPLE
37	S25	85	44	1.667	1.641	0.831	Excellent	INVENTIVE EXAMPLE
38	S26	87	43	1.671	1.623	0.831	Excellent	INVENTIVE EXAMPLE
39	S27	90	44	1.681	1.612	0.834	Excellent	INVENTIVE EXAMPLE
40	S28	85	45	1.603	1.601	0.837	Excellent	INVENTIVE EXAMPLE
41	S29	91	46	1.601	1.601	0.838	Excellent	INVENTIVE EXAMPLE
42	S30	105	44	1.611	1.603	0.836	Excellent	INVENTIVE EXAMPLE
43	S31	80	43	1.650	1.610	0.829	Excellent	INVENTIVE EXAMPLE
44	S32	83	44	1.605	1.601	0.826	Excellent	INVENTIVE EXAMPLE
45	S33	87	43	1.610	1.604	0.824	Excellent	INVENTIVE EXAMPLE
46	S34	79	45	1.625	1.621	0.827	Excellent	INVENTIVE EXAMPLE
47	S35	75	46	1.619	1.606	0.829	Excellent	INVENTIVE EXAMPLE
48	S36	79	43	1.657	1.641	0.828	Excellent	INVENTIVE EXAMPLE
49	S37	82	44	1.661	1.623	0.828	Excellent	INVENTIVE EXAMPLE
50	S38	75	45	1.669	1.612	0.827	Excellent	INVENTIVE EXAMPLE

TABLE 15

No.	STEEL TYPE	MANUFACTURING RESULTS AVERAGE GRAIN SIZE μm	$W_{10/1k}$ W/kg	B_{50} L DIRECTION T	C DIRECTION T	X VALUE	ROUNDNESS	NOTE
51	S39	79	44	1.601	1.601	0.826	Excellent	INVENTIVE EXAMPLE
52	S40	83	45	1.600	1.601	0.827	Excellent	INVENTIVE EXAMPLE
53	S41	77	43	1.602	1.601	0.826	Excellent	INVENTIVE EXAMPLE
54	S42	83	42	1.601	1.601	0.811	Excellent	INVENTIVE EXAMPLE
55	S43	79	43	1.602	1.602	0.815	Excellent	INVENTIVE EXAMPLE
56	S44	73	46	1.603	1.601	0.817	Excellent	INVENTIVE EXAMPLE
57	S45	71	42	1.604	1.602	0.817	Excellent	INVENTIVE EXAMPLE
58	S46	91	41	1.603	1.601	0.819	Excellent	INVENTIVE EXAMPLE
59	S47	88	43	1.602	1.601	0.813	Excellent	INVENTIVE EXAMPLE
60	S48	83	46	1.601	1.601	0.816	Excellent	INVENTIVE EXAMPLE
61	S49	91	44	1.605	1.602	0.807	Excellent	INVENTIVE EXAMPLE
62	S50	81	42	1.604	1.602	0.820	Excellent	INVENTIVE EXAMPLE
63	S51	85	41	1.602	1.601	0.805	Excellent	INVENTIVE EXAMPLE
64	S52	84	39	1.603	1.602	0.811	Excellent	INVENTIVE EXAMPLE
65	S53	65	86	1.743	1.721	0.843	Poor	COMPARATIVE EXAMPLE
66	S54	—	—	—	—	—	—	COMPARATIVE EXAMPLE
67	S55	51	84	1.691	1.641	0.832	Poor	COMPARATIVE EXAMPLE
68	S56	233	82	1.521	1.501	0.798	Poor	COMPARATIVE EXAMPLE
69	S57	—	—	—	—	—	—	COMPARATIVE EXAMPLE
70	S58	23	81	1.634	1.616	0.797	Poor	COMPARATIVE EXAMPLE
71	S59	25	82	1.637	1.623	0.796	Poor	COMPARATIVE EXAMPLE
72	S60	150	59	1.480	1.411	0.781	Poor	COMPARATIVE EXAMPLE
73	S61	—	—	—	—	—	—	COMPARATIVE EXAMPLE
74	S62	15	81	1.510	1.422	0.775	Poor	COMPARATIVE EXAMPLE
75	S63	25	82	1.667	1.633	0.799	Poor	COMPARATIVE EXAMPLE

TABLE 16

No.	STEEL TYPE	MANUFACTURING RESULTS AVERAGE GRAIN SIZE μm	$W_{10/1k}$ W/kg	B_{50} L DIRECTION T	C DIRECTION T	X VALUE	ROUNDNESS	NOTE
76	S64	—	—	—	—	—	—	COMPARATIVE EXAMPLE
77	S65	—	—	—	—	—	—	COMPARATIVE EXAMPLE
78	S66	15	83	1.590	1.580	0.797	Poor	COMPARATIVE EXAMPLE
79	S67	23	81	1.595	1.578	0.798	Poor	COMPARATIVE EXAMPLE
80	S68	25	82	1.597	1.555	0.794	Poor	COMPARATIVE EXAMPLE
81	S69	33	39	1.581	1.541	0.790	Poor	COMPARATIVE EXAMPLE
82	S69	80	43	1.579	1.541	0.789	Poor	COMPARATIVE EXAMPLE
83	S69	187	87	1.649	1.649	0.831	Poor	COMPARATIVE EXAMPLE
84	S70	86	37	1.651	1.621	0.840	Excellent	INVENTIVE EXAMPLE
85	S70	89	69	1.653	1.631	0.842	Excellent	INVENTIVE EXAMPLE
86	S71	89	44	1.531	1.528	0.799	Poor	COMPARATIVE EXAMPLE

INDUSTRIAL APPLICABILITY

[0216] According to the above aspects of the present invention, it is possible to provide the non oriented electrical steel sheet with excellent magnetic characteristics and small mechanical anisotropy for the integrally punched iron core, the iron core, the manufacturing method of the iron core, the motor, and the manufacturing method of the motor. Accordingly, the present invention has significant industrial applicability.

REFERENCE SIGNS LIST

[0217] **1**: NON ORIENTED ELECTRICAL STEEL SHEET

[0218] L: ROLLING DIRECTION

[0219] C: TRANSVERS DIRECTION

1. A non oriented electrical steel sheet comprising a chemical composition containing, by mass %,

0.005% or less of C,

1.0% or more and 5.0% or less of Si,

less than 2.5% of sol. Al,

3.0% or less of Mn,

0.3% or less of P, s

0.01% or less of S,

0.01% or less of N,

0.10% or less of B,

0.10% or less of O,

0.10% or less of Mg,

0.01% or less of Ca,
0.10% or less of Ti,
0.10% or less of V,
5.0% or less of Cr,
5.0% or less of Ni,
5.0% or less of Cu,
0.10% or less of Zr,
0.10% or less of Sn,
0.10% or less of Sb,
0.10% or less of Ce,
0.10% or less of Nd,
0.10% or less of Bi,
0.10% or less of W,
0.10% or less of Mo,
0.10% or less of Nb,
0.10% or less of Y, and
a balance consisting of Fe and impurities, wherein
a sheet thickness is 0.10 mm or more and 0.35 mm or less,
an average grain size is 30 μm or more and 200 μm or less,
an X value defined by a following expression 1 is 0.800 or more, and
an iron loss $W_{10/1k}$ when excited so as to be a magnetic flux density of 1.0 T at a frequency of 1 kHz is 80 W/kg or less,
where the expression 1 is $X=(2\times B_{50L}+B_{50C})/(3\times I_S)$ and
where B_{50L} denotes a magnetic flux density in a rolling direction when magnetized with a magnetizing force of 5000 A/m, B_{50C} denotes a magnetic flux density in a transverse direction when magnetized with a magnetizing force of 5000 A/m, and I_S denotes a spontaneous magnetization at room temperature.

2. The non oriented electrical steel sheet according to claim **1**, wherein the chemical composition contains, by mass %, more than 3.25% and 5.0% or less of Si.

3. The non oriented electrical steel sheet according to claim **1**, wherein the chemical composition contains, by mass %, at least one of
0.0010% or more and 0.005% or less of C,
0.10% or more and less than 2.5% of sol. Al,
0.0010% or more and 3.0% or less of Mn,
0.0010% or more and 0.3% or less of P,
0.0001% or more and 0.01% or less of S,
0.0015% or more and 0.01% or less of N,
0.0001% or more and 0.10% or less of B,
0.0001% or more and 0.10% or less of O,
0.0001% or more and 0.10% or less of Mg,
0.0003% or more and 0.01% or less of Ca,
0.0001% or more and 0.10% or less of Ti,
0.0001% or more and 0.10% or less of V,
0.0010% or more and 5.0% or less of Cr,
0.0010% or more and 5.0% or less of Ni,
0.0010% or more and 5.0% or less of Cu,
0.0002% or more and 0.10% or less of Zr,
0.0010% or more and 0.10% or less of Sn,
0.0010% or more and 0.10% or less of Sb,
0.001% or more and 0.10% or less of Ce,
0.002% or more and 0.10% or less of Nd,
0.002% or more and 0.10% or less of Bi,
0.002% or more and 0.10% or less of W,
0.002% or more and 0.10% or less of Mo,
0.0001% or more and 0.10% or less of Nb, and
0.002% or more and 0.10% or less of Y.

4. The non oriented electrical steel sheet according to claim **1**, wherein the chemical composition contains, by mass %, more than 4.0% in total of Si and sol. Al.

5. The non oriented electrical steel sheet according to claim **1**, wherein the X value is 0.800 or more and less than 0.845.

6. The non oriented electrical steel sheet according to claim **1**, wherein the X value is 0.800 or more and less than 0.830.

7. An iron core comprising the non oriented electrical steel sheet according to claim **1**.

8. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim **1**.

9. A motor comprising the iron core according to claim **7**.

10. A manufacturing method of a motor comprising
a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim **1** and
a process of assembling the motor using the iron core.

11. An iron core comprising the non oriented electrical steel sheet according to claim **2**.

12. An iron core comprising the non oriented electrical steel sheet according to claim **3**.

13. An iron core comprising the non oriented electrical steel sheet according to claim **4**.

14. An iron core comprising the non oriented electrical steel sheet according to claim **5**.

15. An iron core comprising the non oriented electrical steel sheet according to claim **6**.

16. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim **2**.

17. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim **3**.

18. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim **4**.

19. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim **5**.

20. A manufacturing method of an iron core comprising a process of punching and laminating the non oriented electrical steel sheet according to claim **6**.

21. A manufacturing method of a motor comprising
a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim **2** and
a process of assembling the motor using the iron core.

22. A manufacturing method of a motor comprising
a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim **3** and
a process of assembling the motor using the iron core.

23. A manufacturing method of a motor comprising
a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim **4** and
a process of assembling the motor using the iron core.

24. A manufacturing method of a motor comprising
a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim **5** and
a process of assembling the motor using the iron core.

25. A manufacturing method of a motor comprising
a process of preparing an iron core by punching and laminating the non oriented electrical steel sheet according to claim **6** and
a process of assembling the motor using the iron core.

26. A non oriented electrical steel sheet comprising a chemical composition containing, by mass %,
0.005% or less of C,
1.0% or more and 5.0% or less of Si,
less than 2.5% of sol. Al,
3.0% or less of Mn,
0.3% or less of P,
0.01% or less of S,
0.01% or less of N,
0.10% or less of B,
0.10% or less of O,
0.10% or less of Mg,
0.01% or less of Ca,
0.10% or less of Ti,
0.10% or less of V,
5.0% or less of Cr,
5.0% or less of Ni,
5.0% or less of Cu,
0.10% or less of Zr,
0.10% or less of Sn,
0.10% or less of Sb,
0.10% or less of Ce,
0.10% or less of Nd,
0.10% or less of Bi,
0.10% or less of W,
0.10% or less of Mo,
0.10% or less of Nb,
0.10% or less of Y, and
a balance comprising Fe and impurities, wherein
a sheet thickness is 0.10 mm or more and 0.35 mm or less,
an average grain size is 30 μm or more and 200 μm or less,
an X value defined by a following expression 1 is 0.800 or more, and
an iron loss $W_{10/1k}$ when excited so as to be a magnetic flux density of 1.0 T at a frequency of 1 kHz is 80 W/kg or less,
where the expression 1 is $X=(2\times B_{50L}+B_{50C})/(3\times I_S)$ and
where B_{50L} denotes a magnetic flux density in a rolling direction when magnetized with a magnetizing force of 5000 A/m, B_{50C} denotes a magnetic flux density in a transverse direction when magnetized with a magnetizing force of 5000 A/m, and I_S denotes a spontaneous magnetization at room temperature.

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